How-To Library

Informatica

Deploying the Informatica® PowerCenter 10.4.0 AWS Marketplace Solution

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Abstract

This deployment reference provides step-by-step instructions for deploying Informatica® PowerCenter on the Amazon Web Services Cloud Marketplace. Automated reference deployments use AWS CloudFormation templates to launch, configure, and run the AWS compute, network, storage, and other services required to deploy a specific workload on AWS.

Supported Versions

- PowerCenter 10.4.0

Table of Contents

Overview . . . 2
Intended Audience . . . 3
Cost and Licenses . . . 3
Architecture . . . 3
AWS Resources Created by the Deployment . . . 4
Informatica domain . . . 6
Informatica Clients . . . 6
Before You Begin . . . 7
Specialized Knowledge . . . 7
Network Prerequisites . . . 7
Account and Security Prerequisites . . . 8
Storage Prerequisites . . . 9
Deploying PowerCenter on the AWS Marketplace . . . 9
Step 1. Get a License for PowerCenter . . . 10
Step 2. Launch the Deployment . . . 10
Step 2. Specify Stack Details . . . 11
Monitoring Instance Provision and Informatica Domain Creation . . . 16
Deployed AWS Resources . . . 16
Output Tab Properties . . . 17
Deployed Informatica Resources . . . 18
Solution Deployment Logs . . . 19
Access Domain in the AWS Cloud from PowerCenter Clients . . . 20

Overview

PowerCenter is a data integration product that transforms fragmented, raw data from any source, at any latency into complete, actionable information. You can integrate data from on-premises data sources, cloud data sources, or Amazon Web Services (AWS) with PowerCenter.

Customers of Amazon Web Services and Informatica can launch a PowerCenter 10.4.0 deployment from the AWS marketplace to create an Informatica domain in the AWS cloud and explore PowerCenter functionality.

Intended Audience

This guide is for users responsible for deploying the marketplace solution of PowerCenter 10.4.0 on the AWS Cloud.

As a user with administrator privileges to deploy applications on AWS, you should be familiar with AWS resources such as Cloud formation, VPC, EC2 instances, S3 storage, RDS, Internet gateway, NAT Gateway, route tables, security groups and roles, and Elastic IP. See the AWS documentation.

Familiarity with the concepts of IP CIDR, public and private IP addresses is recommended.

You should also be familiar with PowerCenter. To find PowerCenter documentation, see the Informatica documentation portal.

Cost and Licenses

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this marketplace deployment.

The AWS CloudFormation template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. See the pricing pages for each AWS service you will be using for cost estimates.

This deployment requires a license for Informatica PowerCenter. To sign up for a demo license, contact your organization's Informatica sales contact or visit Informatica Global Customer Support.

The following table lists the instance types that you can choose based on sizing requirements:

Virtual Machine	Instance Type
Database	m5.xlarge
Informatica Domain	m5.2xlarge
Bastion Server	m5.xlarge

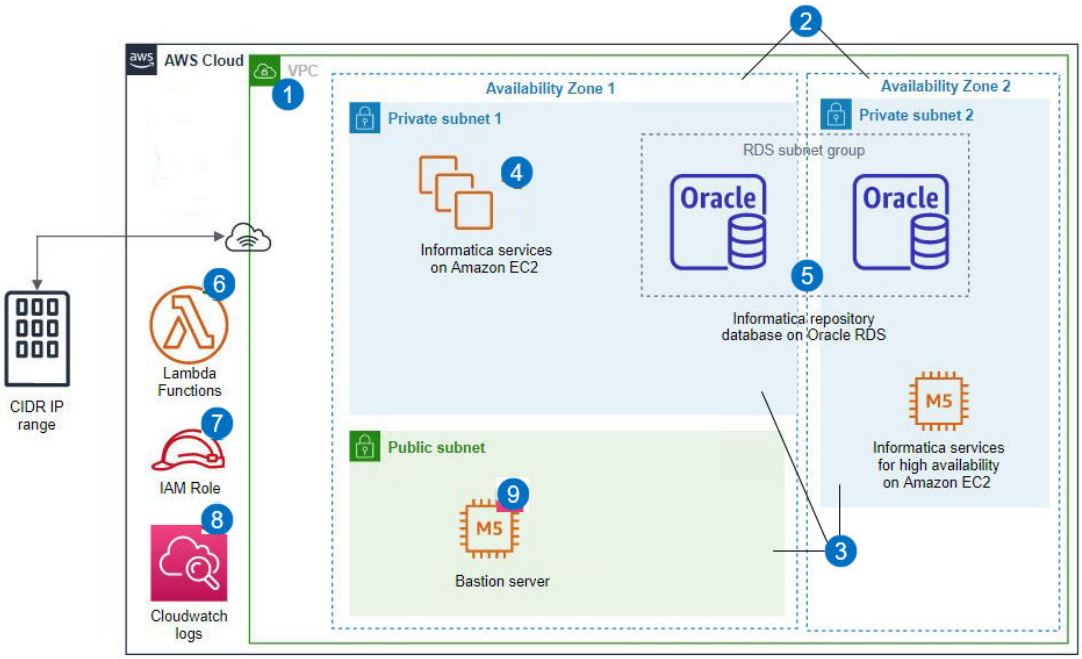
The deployment chooses M5 type of instances based on the instance type availability for the specific AWS region.

Architecture

When you deploy PowerCenter marketplace solution in a new VPC, AWS CloudFormation templates create and connect the following resources in the VPC:

- Informatica domain on an EC2 instance, with additional instances to contain nodes in the PowerCenter grid.
- PowerCenter clients on a remote Windows bastion server, on a public or private subnet.
- Amazon RDS relational databases for Informatica repositories.
- AWS security and account management services.

The following image shows the PowerCenter deployment for new VPC on AWS architecture with the Amazon Marketplace:



The icons in the architecture diagram correspond to items in the following list:

1. A virtual public cloud (VPC) configured across two Availability Zones to contain the PowerCenter deployment.
2. Availability zones.
3. Subnets to contain specific elements of the deployment. Creates two private subnets, plus one public subnet if you want to use a Windows bastion server for Informatica clients. Create each of the subnets in two availability zone.
4. Informatica domain.
5. Oracle database to contain the following Informatica repositories:
   - PowerCenter repository
   - Domain configuration repository
6. AWS Lambda functions.
7. IAM roles.
8. Amazon CloudWatch.
9. Informatica clients in a separate EC2 bastion server in a public subnet.

AWS Resources Created by the Deployment

The deployment process creates or includes the following components:

- Components created when you specify the solution to create a new VPC.
- Components created, as well as some that already exist, when you specify the solution to use an existing VPC.
- A bastion server, if you choose to create one.

Components in a Deployment on a New VPC

The following additional components will be created when you deploy the software in a new virtual private cloud (VPC) with the components in Existing VPC:

Component	Number of Components Created
VPC	1
Internet gateway	1
Subnet	4
Route table	2
Security group	The following security groups: - One security group for the RDS - One security group for the Informatica PowerCenter instance - If you selected the option to create the bastion server, one security group to support remote login to the bastion server.
DB subnet group	1
RDS	1 Oracle RDS instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type.
EC2	Based on your choice, one to six Informatica node with an EC2 instance and 200 GB(150 GB for windows node) of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice one of the above mentioned node will be used as high availability node. Based on your choice, a bastion server with an EC2 instance that has 150 GB of storage space is created. The instance created is of type m5.large.
IAM role and Instance Profile	1
Elastic IP	1 Elastic IP attached to the NAT gateway.
EFS File System	1

Components in a Deployment on an Existing VPC

The following table lists components that the cloud formation template creates when you deploy the software in an existing VPC:

Component	Number of Components Created
Security group	The following security groups: - One security group for the RDS - One security group for the Informatica PowerCenter instance - If you selected the option to create the bastion server, one security group to support remote login to the bastion server.
DB subnet group	1

Component	Number of Components Created
RDS	1 Oracle RDS instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type.
EC2	Based on your choice, one to six Informatica node with an EC2 instance and 200 GB(150 GB for windows node) of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice one of the above mentioned node will be used as high availability node. Based on your choice, a bastion server with an EC2 instance that has 150 GB of storage space is created. The instance created is of type m5.large.
AWS Lambda Function	5 - Lambda functions to check the feasability for VPC , Subnets and Informatica License
IAM role and Instance Profile	1
EFS File System	1

Bastion Server

You can optionally deploy a bastion server as an element in the solution. The bastion server acts as a firewall between the internet and the cloud platform network where the solution deploys. It can also act as a remote server which you can log into to run Informatica clients.

Informatica domain

The Informatica domain is a server component that hosts application services, such as PowerCenter Repository Service and PowerCenter Integration Service. You can choose to create PowerCenter Grid depending on the requirement. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

PowerCenter Repository Service

The PowerCenter Repository Service manages the PowerCenter repository. The PowerCenter repository stores metadata created by Informatica products in a relational database to enable collaboration among the products.

PowerCenter Integration Service

The PowerCenter Integration Service is an application service in the Informatica domain that performs data integration tasks.

Informatica Clients

You can use several different clients with PowerCenter:

Administrator tool

The Administrator tool enables you to create and administer services.

PowerCenter clients

The PowerCenter Client consists of the following applications that you can use to design mappings, mapplets, create sessions and workflows to load the data, and monitor workflow progress:

- PowerCenter Designer

- PowerCenter Workflow Monitor
- PowerCenter Repository Manager
- PowerCenter Workflow Manager

Bastion server

You can optionally deploy a bastion server as an element in the solution. The bastion server is an AWS Windows instance installed with the Developer tool and the command line interface clients. The bastion server acts as a firewall for access to the VPC. It can also act as a remote Windows server that you can log into to run Informatica clients.

Before You Begin

Before you deploy PowerCenter in the AWS cloud, verify that you have performed the following steps:

- Complete all the network prerequisites.
- Verify the user permissions.
- Verify the storage prerequisites.

Specialized Knowledge

Before you deploy PowerCenter, we recommend that you become familiar with the following AWS services.

If you are new to AWS, see Getting Started with AWS:

- Amazon VPC
- Amazon EC2
- Amazon RDS
- Amazon S3
- Elastic IP Addresses

If you are not familiar with PowerCenter and the Informatica platform, see Informatica documentation portal.

Network Prerequisites

Before you deploy PowerCenter in the AWS cloud, verify that you have completed the following network prerequisites for PowerCenter.

Network prerequisites depend on whether you choose to deploy the solution to a new subnet or an existing subnet.

When you deploy into a new VPC, the automated deployment builds a new AWS environment consisting of the VPC, subnets, NAT gateways, security groups, and other infrastructure components, and then deploys PowerCenter into this new VPC. When you deploy to an existing VPC, the automated deployment provisions PowerCenter in your existing AWS infrastructure.

New VPC Prerequisites

Verify the following requirements for all VPCs and subnets:

- Ensure that the selected availability zone has sufficient capacity to create a new subnet, route table, internet gateway, and NAT gateway.
- Ensure that the VPC creation has not exceeded the supported limit.

Existing VPC Prerequisites

In addition, observe the following prerequisites when you choose to deploy into an existing VPC:

- Ensure that the VPC has DNS resolution enabled.
- You must also attach the subnet to a route table with a local route to the VPC CIDR.
- Verify that each subnet in the VPC is associated with a route table.
- Verify that the subnet associated with an internet gateway has the Auto-assign Public IPv4 property enabled.
- Verify that each subnet, and each route listed in the route table, is defined as one of the following gateways:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- For a successful deployment, verify that the service subnets you choose are attached to a route table under the selected VPC.
- Verify that your VPC has a minimum of two subnets. If you choose to create a bastion server, you can deploy it to one of the two subnets or use a separate public subnet.

Subnets require an Internet gateway, an NAT gateway, or a virtual private gateway in their route tables, to allow the instances to download packages and software without exposing them to the Internet. Record the domain name option configured in the DHCP options as explained in the Amazon VPC documentation. You provide the name in VPC settings when you configure the deployment.

Account and Security Prerequisites

Observe the prerequisites in this section to verify that you established the required AWS account and the requisite account private keys.

AWS Account and Region

If you don't already have an AWS account, create one at aws.amazon.com by following the on-screen instructions.

Use the region selector in the navigation bar to choose the AWS region where you want to deploy PowerCenter on AWS.

Note: The region you choose determines the availability of certain sizes of EC2 instances.

If necessary, request a service limit increase for the EC2 instance type you've decided to use for the Informatica domain. You might need to do this if you already have an existing deployment that uses this instance type, and you think you might exceed the default limit with this reference deployment.

Private Keys

Create a key pair in your preferred region.

When you log in to an EC2 instance, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.

If you do not have an existing .pem key to use, follow the instructions in the AWS documentation to create a key pair.

Note: Your administrator might ask you to use an existing key pair.

When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the PowerCenter deployment, so that you can provide the key pair name during network configuration.

Account User Policies

Choose to grant the AWS user one of the following:

- Administrator privileges
- The following minimum policies attached to the user account:
  - arn:aws:iam::aws:policy/AWSCloudFormationFullAccess
  - arn:aws:iam::aws:policy/AmazonEC2FullAccess
  - arn:aws:iam::aws:policy/CloudWatchFullAccess
  - arn:aws:iam::aws:policy/IAMFullAccess
  - arn:aws:iam::aws:policy/AmazonS3FullAccess
  - arn:aws:iam::aws:policy/AmazonRDSFullAccess
  - arn:aws:iam::aws:policy/AmazonVPCFullAccess
  - arn:aws:iam::aws:policy/AWSLambdaFullAccess
  - arn:aws:iam::aws:policy/AmazonElasticFileSystemFullAccess

Storage Prerequisites

The PowerCenter license must be stored in an Amazon S3 bucket.

Amazon S3 (Simple Storage Service) is an object repository in AWS. S3 stores objects, consisting of one or more files and accompanying file metadata, in "buckets." For information about S3, see the AWS documentation.

To use the automated cloud marketplace solution, you must create an S3 bucket to store the PowerCenter license. You can store the license on the bucket level or in a folder within the bucket.

Note the location of the license key and supply it when you configure the solution in the **License Key S3 Bucket URI** parameter.

Deploying PowerCenter on the AWS Marketplace

The automated deployment of PowerCenter on the AWS marketplace uses the AWS CloudFormation console to guide your choices and launch the solution deployment.

Use the AWS CloudFormation console to specify the stack name and to enter parameter values to the template that you use to deploy the application. A stack is a logical representation of an AWS deployment.

Specify the identifiers for the VPC, for the subnet that contains the Amazon EC2 instances where the deployment process deploys the application components, and for the subnet where the deployment process creates the PowerCenter repository. The deployment process provisions the application components and creates the PowerCenter repository based on the values that you specify.

Availability Zones and Subnets

The automated Amazon Marketplace solution deploys the components and services for PowerCenter into an Amazon VPC configured across two Availability Zones. Each Availability Zone is provisioned with two subnets.

In the first subnet, the deployment process provisions PowerCenter application services on an Amazon EC2 instance within the VPC:

- PowerCenter Repository Service
- PowerCenter Integration Service

For more information about domain components, see the Informatica Administrator Guide and Informatica Application Service Guide.

In the second subnet, the solution deploys an Oracle database on Amazon RDS for use as the PowerCenter repository. The PowerCenter repository stores metadata created by Informatica services.

Step 1. Get a License for PowerCenter

1. This deployment requires a license for Informatica PowerCenter. To sign up for a demo license, please contact Informatica.
2. Place the PowerCenter license key file for the software in the S3 bucket or directory that you created. You'll be prompted for its location in a later step.

Step 2. Launch the Deployment

You can configure PowerCenter in Amazon Marketplace for Windows or Linux as a Bring Your Own License (BYOL) listing.

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this deployment. See the pricing pages for each AWS service you will be using for full details. Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Login to AWS Marketplace using the AWS account to configure PowerCenter for Windows or for Red Hat Linux (BYOL) based on the operating system that you want to use.
2. Type `Informatica PowerCenter` in the search box and press **Enter**.
3. Click **Informatica PowerCenter 10.4.0**.

   The **Product Overview** page appears.
4. Click **Continue to Subscribe**.

   The **Subscribe to this software** page appears.
5. Click **Continue to Configuration**.

   The **Configure this software** page appears.
6. Choose whether to deploy the solution on a new VPC, existing VPC, or domain expansion.
7. Click **Continue to Launch**.

   The **Launch this software** page appears.
8. Select **Launch CloudFormation** from the **Choose Action** drop-down list and click **Launch**.

   The **Create stack** page appears.
9. Verify the region for the deployment. The console displays the region in the upper-right corner of the navigation bar. Change the region if necessary.
10. In the Stack pane, ensure that the default option `Amazon S3 URL` is selected in the **Specify Template** section.

    Caution: Do not change any settings on this page or navigate away from the page. If you leave the page, the Amazon S3 URL might be erased.
11. Click **Next**.

    The **Specify Stack Details** page appears.

Step 2. Specify Stack Details

On the **Specify stack details** page, review the parameters for the template. Provide values for the parameters that require input. For all other parameters, review the default settings and customize them as necessary.

1. Change the stack name if needed.
2. Supply values for the parameters in the **CloudFormation** screen.

   You can use the information in the sections below to supply the values.
3. Under **Capabilities**, select the check box to acknowledge that the template will create IAM resources
4. To deploy the stack, click **Create**.

Amazon AWS begins to create the stack and displays the CloudFormation dashboard. When the status is CREATE_COMPLETE, the deployment is complete. You can use the URL displayed on the **Output tab** to view the stack properties.

Parameters to Deploy to a New VPC

When you choose to deploy the PowerCenter solution to a new VPC, you supply the following parameters on the **Specify Stack Details** page. Parameters marked with an asterisk (*) are mandatory.

Stack name

Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Network Configuration

Enter values for the following parameters:

Parameter	Default	Description
Availability Zones*	Requires selections	List of Availability Zones to use for the subnets in the VPC. Multi-select in the dropdown list to choose at least two zones.
VPC CIDR*	Requires input	CIDR block for the new VPC. Format: `x.x.x.x/x` For example: `10.0.0.20/16`
Public Subnet CIDR*	Requires input	CIDR block for the public subnet. Must be within the VPC CIDR.
Private Subnet 1 CIDR*	Requires input	CIDR block for private subnet 1. Must be within the VPC CIDR.
Private Subnet 2 CIDR*	Requires input	CIDR block for private subnet 2. Must be within the VPC CIDR.
CIDR IP Range*	Requires input	The CIDR IP range that is permitted to access the Informatica domain. We recommend that you use a constrained CIDR range to reduce the potential of inbound attacks from unknown IP addresses. For example, to specify the range of 10.20.30.40 to 10.20.30.49, enter 10.20.30.40/49.

Bastion Server for Informatica

Enter values for the following parameters:

Parameter	Default	Description
Deploy Bastion Server	No	Select Yes to deploy a bastion server to permit external clients to access resources in the VPC.

Amazon EC2 Configuration

Enter values for the following parameters:

Parameter	Default	Description
Key Pair Name*	Requires input	Select an EC2 key pair to enable SSH and RDP access to the PowerCenter solution.

Amazon Elastic File System (EFS) Configuration (applicable only on Linux deployment)

Enter values for the following parameters:

Parameter	Default	Description
EFS Creation	No	Select Yes to create an Amazon Elastic File System.

Informatica PowerCenter Configuration

Enter values for the following parameters:

Parameter	Default	Description
License Key S3 URI*	Requires input	Enter the S3 Bucket URI for the PowerCenter license key. Use the following format: `s3://bucket-name/license.key` or `s3://bucket-name/sub-folder/license.key`
High Availability Deployment	Disable	Choose to enable or disable high availability for the PowerCenter deployment. For information about high availability for the Informatica domain, see the Informatica documentation.
Number of PowerCenter Nodes*	Requires input	Select the number of nodes that you need to run PowerCenter.

Parameters to Deploy to an Existing VPC

When you choose to deploy the PowerCenter solution to an existing VPC, you supply the following parameters on the **Specify Stack Details** page. Parameters marked with an asterisk (*) are mandatory.

Stack name

Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Network Configuration

Enter values for the following parameters:

Parameter	Default	Description
VPC*	Requires input	The identifier for the existing VPC
Subnet 1*	Requires input	The identifier for the subnet within the VPC where you want to deploy PowerCenter.
Subnet 2*	Requires input when High Availability is enabled.	The identifier for the subnet used by PowerCenter when High Availability is enabled. **Note:** This subnet must be in a different Availability Zone than Subnet 1.
CIDR IP Range*	Requires input	The CIDR IP range that is permitted to access the Informatica domain. We recommend that you use a constrained CIDR range to reduce the potential of inbound attacks from unknown IP addresses. For example, to specify the range of 10.20.30.20 to 10.20.30.24, enter 10.20.30.20/24.
Pre-validation Check	No	Validates the VPC and subnets before deploying the PowerCenter solution. If you choose to have the automated solution perform a validation check before launching the deployment, the check validates your parameter inputs. If the deployment as specified fails the check, the deployment process stops. The deployment logs record the problem.

Bastion Server for Informatica

Enter values for the following parameters:

Parameter	Default	Description
Deploy Bastion Server	No	Select Yes to deploy a bastion server to permit external clients to access resources in the VPC.
Bastion Subnet*	Requires input if you choose to deploy a bastion server	Select a subnet for the bastion server. The subnet must belong to the VPC that you specified for the bastion server. **Tip:** Choose a subnet that is attached to an internet gateway.

Amazon EC2 Configuration

Enter values for the following parameters:

Parameter	Default	Description
Key Pair Name*	Requires input	Select an EC2 key pair to enable SSH and RDP access to the PowerCenter solution.

Amazon Elastic File System (EFS) Configuration (applicable only on Linux deployment)

Enter values for the following parameters:

Parameter	Default	Description
EFS Creation	No	Select Yes to create an Amazon Elastic File System.

Informatica PowerCenter Configuration

Enter values for the following parameters:

Parameter	Default	Description
License Key S3 Bucket URI*	Requires input	Enter the S3 Bucket URI for the PowerCenter license key. Use the following format: `s3://bucket-name/license.key` or `s3://bucket-name/sub-folder/license.key`
High Availability Deployment	Disable	Choose to enable or disable high availability for the PowerCenter deployment. For information about high availability for the Informatica domain, see the Informatica documentation.
Number of PowerCenter Nodes*	Requires input	Select the number of nodes that you need to run PowerCenter.

Parameters to Deploy with Domain Expansion

When you choose to deploy the PowerCenter solution to an existing VPC, you supply the following parameters on the **Specify Stack Details** page. Parameters marked with an asterisk (*) are mandatory.

Stack name

Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Network Configuration

Enter values for the following parameters:

Parameter	Default	Description
VPC*	Requires input	The identifier for the existing VPC
Subnet*	Requires input	Select the publicly accessible subnet ID located in Availability Zone for the Informatica domain. It is the same as the subnet for the Informatica Master Node.
Security Group*	Requires input	Select a security group that belongs to the public subnet created at the time of domain deployment.

Amazon EC2 Configuration

Enter values for the following parameters:

Parameter	Default	Description
Key Pair Name*	Requires input	Select an EC2 key pair to enable SSH and RDP access to the PowerCenter solution.
Instance Role	Requires input	Enter the instance role created at the time of domain deployment.
Instance Profile*	Requires input	Enter the instance profile created at the time of domain deployment.

Amazon Elastic File System (EFS) Configuration (applicable only on Linux deployment)

Enter values for the following parameters:

Parameter	Default	Description
EFS Created*	No	Select Yes to mount the existing Amazon Elastic File System.
EFS File System Endpoint ID*	Requires input	Enter the EFS ID for the subnet created at the time of domain deployment.

RDS Details

Enter values for the following parameters:

Parameter	Default	Description
RDS Endpoint*	Requires input	Enter the RDS endpoint address of the Informatica domain database.

Informatica PowerCenter Configuration

Enter values for the following parameters:

Parameter	Default	Description
Gateway Node Private IP*	Requires input	Enter the private IP address of the gateway node.
Informatica Administrator User*	Requires input	Enter the user name of the administrator account for Informatica PowerCenter. Default is Administrator.
Informatica Administrator Password*	Requires input	Enter the administrator password for accessing PowerCenter. The string is 8-30 characters long, with at least one special character, one number, one uppercase character, and one lowercase character. You cannot specify double quotes ("), ampersands (&), and dollar signs ($).
Node Name*	Requires input	Enter a name for the new node to add to the Informatica domain.

Monitoring Instance Provision and Informatica Domain Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain on the cloud platform.

Deployed AWS Resources

You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal

Each row of the display shows stack components and creation status. The stack is complete when the status reads CREATE_COMPLETE.

For deployment to a new VPC, the console displays the following stack components:

Component	Description
StackName-VPCStack	When you deploy to a new VPC, this template creates network-related resources, including the VPC, subnets, internet gateway, and the NAT gateway.
PCStack	The main stack.

In addition to the above components, the console displays the following stack components when you deploy to an existing VPC:

Component	Description
StackName/PCStack	Main stack. Always created.
StackName-LicenseCheckLambdaStack	Used to check if provided license key file is valid. Created by default.
StackName-BastionSubnetCheckLambdaStack	Used to check if a subnet is valid. Created by default. Lambda function will be created only when the following parameters are set to Yes: - **Pre-validation Check** - **Deploy Bastion Server**
StackName-SubnetsAZCheckLambdaStack	Used to check if a subnets are in different availability zones. Created by default.
StackName-PublicIpDnsCheckLambdaStack	Used to check if an auto-assign public IPv4 address is enabled for subnet. Created by default.
StackName-Subnet2ChecklambdaStack	Used to check if a subnet is valid. Default. Lambda function will be created if the **Pre-validation Check** parameter is set to Yes.
StackName-Subnet1CheckLambdaStack	Used to check if a subnet is valid. Default. Lambda function will be created if the **Pre-validation Check** parameter is set to Yes.

Component	Description
StackName-IAMStack	Creates IAM roles and profiles used by EC2 instances. Created by default.
StackName-EFSStack	Creates Elastic File system and its endpoint. Created only when the **Create EFS** parameter set to `Yes` and the region supports it. **Note:** Applicable for Linux deployment.
StackName-SecurityGroupStack	Creates security groups and network interfaces used by EC2 instances. Created by default.
StackName-RDSStack	Creates RDS resources. Created by default.
StackName-BastionServerStack	Creates bastion server on an EC2 instance. Created only when you choose to create the bastion server.
StackName-PCServerStack	Creates the Informatica PowerCenter gateway node EC2 instance. Created by default.
StackName-PCNode2Stack	Creates the Informatica PowerCenter node2 or HA node EC2 instance. Created only when there are two or more PowerCenter nodes or when **High Availability** is enabled.
StackName-PCNode3Stack	Creates the Informatica PowerCenter node3 EC2 instance. Created only when there are three or more PowerCenter nodes.
StackName-PCNode4Stack	Creates the Informatica PowerCenter node4 EC2 instance. Created only when there are four or more PowerCenter nodes.
StackName-PCNode5Stack	Creates the Informatica PowerCenter node5 EC2 instance. Created only when there are five or more PowerCenter nodes.
StackName-PCNode6Stack	Creates the Informatica PowerCenter node6 EC2 instance. Created only when there are six or more PowerCenter nodes.

Output Tab Properties

Click the **Output** tab to view the list of resources that the deployment process created in the VPC. The table below lists the outputs:

Output	Description
AdminConsoleUser	User ID of the Administrator tool client user.
AdminConsoleUserPwd	Password for the Administrator tool user.
BastionPrivateDNS	DNS of the private bastion server. For example: ip-<IPv4 address>-<region designation>.compute.internal

Output	Description
BastionPublicIPDNS	DNS of the public bastion server. For example: ip-<IPv4 address>-<region designation>.compute.amazonaws.com
DBUserPwd	Password for the database users of the domain databases created in the deployment.
DBUsers	User IDs for the database users of the domain databases created in the deployment.
HANodeIP	IPv4 address of the high availability or failover node for the domain.
InformaticaAdminConsolePrivateURL	URL:port of the Administrator tool hosted on a private server.
InformaticaAdminConsolePublicURL	Public URL:port of the Administrator tool.
InformaticaAdminConsoleServerLog	Path to the domain installation log.
InstanceSetupLogs	Path to the domain EC2 host setup log.

The table below lists the outputs for domain expansion:

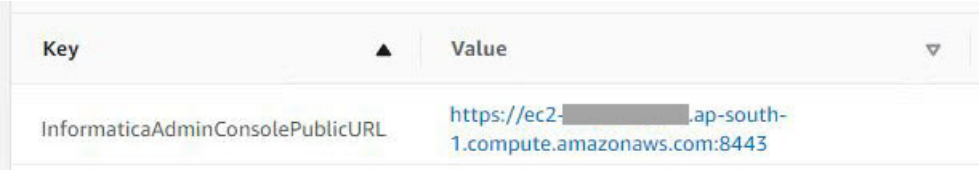
Key	Description
NodePrivateIP	Informatica Joined Node Private IP.

Use the URL, user ID, and password to access the Administrator tool to perform administrator tasks. For a description of domain administrator tasks, see the Informatica Administrator Guide and the Data Engineering Integration Administrator Guide.

Deployed Informatica Resources

The list of deployed resources includes the **InformaticaAdminConsolePublicURL**. Click the linked value of this property.

The following image shows an example of this property, with the IP address portion redacted:



The Administrator tool opens in a browser window.

Use the Informatica Administrator user name and password that are listed in the **Outputs** list to log in to the Administrator tool.

You can use the Administrator tool to view and administer the Informatica application services and resources that the automated deployment created.

Solution Deployment Logs

You can review the following logs to see the solution deployment as it happened. Review logs to see the success or failure of solution element creation.

Review the following logs in the `/var/log` directory in Linux deployment and `C:\cfn\log\` directory in Windows deployment:

- *cfn-init.log*
- *cfn-init-cmd.log*
- Available only in Linux deployment.*cloud-init.log*

Review the following log in the `/opt` directory in Linux deployment and `C:\` directory in Windows deployment :

- Infa_OneClick_Solution.log

Informatica Domain Logs

You can access the following logs on the Informatica domain VM:

Command execution log

This log records the following events:

- Creation of Informatica connections, cluster configurations, and services.
- Population of the data warehouse and SQL databases.
- PowerCenter Integration Service recycling to register all changes.

At the top of the log file is a summary section that lists automated tasks and their status. Beneath the summary section are detailed sections about each task. If any of the tasks failed to complete successfully, you can look at the detailed section for the task to troubleshoot the task.

File name: *Infa_OneClick_Solution.log*

Linux Location: `/opt`

Windows Location: `C:\`

Linux Location: `/opt`

File name: *cfn-init-cmd.log*

Windows Location: `C:\cfn\log\`

Linux Location: `/var/log`

File name: *cfn-init.log*

Windows Location: `C:\cfn\log\`

File name: *cloud-init.log*

Linux Location: `/var/log`

CloudWatch logs for Informatica

CloudWatch is an AWS service that enables you to monitor all of the events in your AWS deployments. CloudWatch provides a central location from which to monitor EC2 instances and other resources. for more information, see the AWS documentation.

To monitor PowerCenter events in CloudWatch, go to the CloudWatch page and search for "Informatica."

If no log files are returned by the search, log into the VM where the Informatica domain is installed and check for logs with the filename "`cfn*.log`"

Access the Informatica deployment logs through the AWS console:

1. In the Amazon Web Services Console, select CloudWatch from the Services menu.
2. In the navigation pane, choose **Logs** > **Log groups**.
3. Enter `InformaticaPCLogGroup` in the search pane.
4. Select one of the following logs based on your operating system to troubleshoot:
   - Linux. installationLogs, cfn-init
   - Windows. installationLogsWindows, cfn-init-windows

CloudWatch logs for Lambda functions

If the automated deployment fails during execution, CloudWatch logs display information about process errors.

Access the CloudWatch logs through the AWS console:

1. In the Amazon Web Services Console, select CloudWatch from the Services menu.
2. In the navigation pane, choose **Logs** > **Log groups**.
3. Enter `/aws/lambda/<stack name>` in the search pane.
4. Select the log group associated with the name of the stack that failed.

Access Domain in the AWS Cloud from PowerCenter Clients

Connect to the domain in the AWS cloud irrespective of the geographic region.

1. In the AWS **CloudFormation** console, select the main stack.
2. To access the Administrator tool in a browser, enter one of the following values from the output tab:
   - InformaticaAdminConsolePublicURL
   - InformaticaAdminConsolePrivateURL
3. Enter the domain server IP address in the client host file.

   Ensure that the public IP maps to the host name.

   Windows. `C:\Windows\System32\drivers\etc`

   Linux. `/etc/hosts`
4. Save and close the file.
5. Log in to the PowerCenter clients and connect to the domain.

   Note: If you have deployed PowerCenter with Bastion Server, log in to the Bastion Server to directly connect to the PowerCenter clients and domain.

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